GIT AND GITHUB TASK 2

1.Install git.

Download Git

Run the installer

Open Git Bash

Verify Git installation

```
Shubhasri@LAPTOP-LK3KFERI MINGW64 /
$ git --version
git version 2.37.1.windows.1

Shubhasri@LAPTOP-LK3KFERI MINGW64 /
$ git config --global --list
user.email=shubhasri015@gmail.com
user.name=shubhasri015-hash

Shubhasri@LAPTOP-LK3KFERI MINGW64 /
$ git config --global user.name "shubhasri015-hash"

Shubhasri@LAPTOP-LK3KFERI MINGW64 /
$ git config --global user.email "shubhasri015@gmail.com"

Shubhasri@LAPTOP-LK3KFERI MINGW64 /
$
```

2. Create a repo in github with README.md and .ignore file.

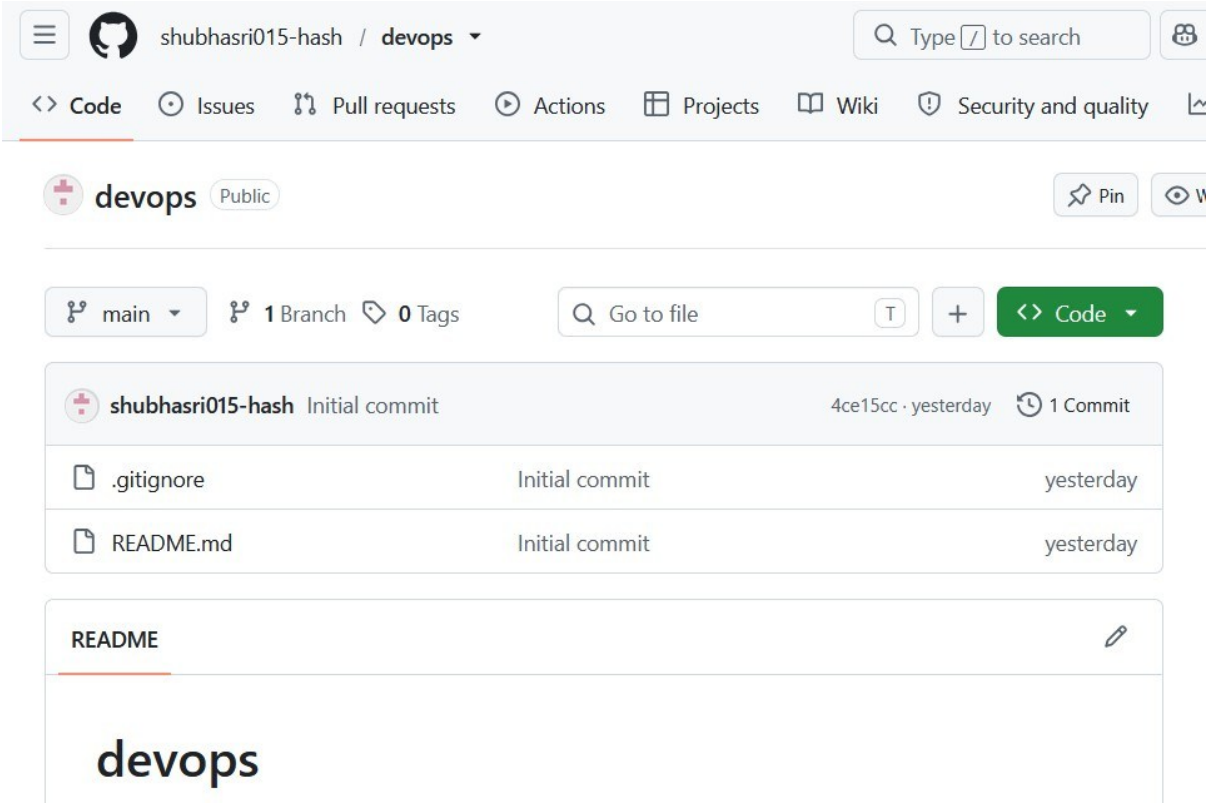
```
Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads (master)
$ git clone git@github.com:shubhasri015-hash/devops.git
Cloning into 'devops'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), done.

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads (master)
$ cd devOps

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devOps (main)
$ ls -a
./  ../  .git/  .gitignore  README.md

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devOps (main)
$
```

Create a new repository , Check Add a README file., Under Add .gitignore, Click Create repository.



GitHub will create your repository with README.md , Create the .gitignore file , 🗄 Open your repository.

Click Add file → Create new file. Enter the filename , Click Commit changes.

3. Clone the created repo to local.

Open Git Bash -Go to the location where you want to save your repository.- Click the green Code button and copy the ssh URl and paste in gitbash

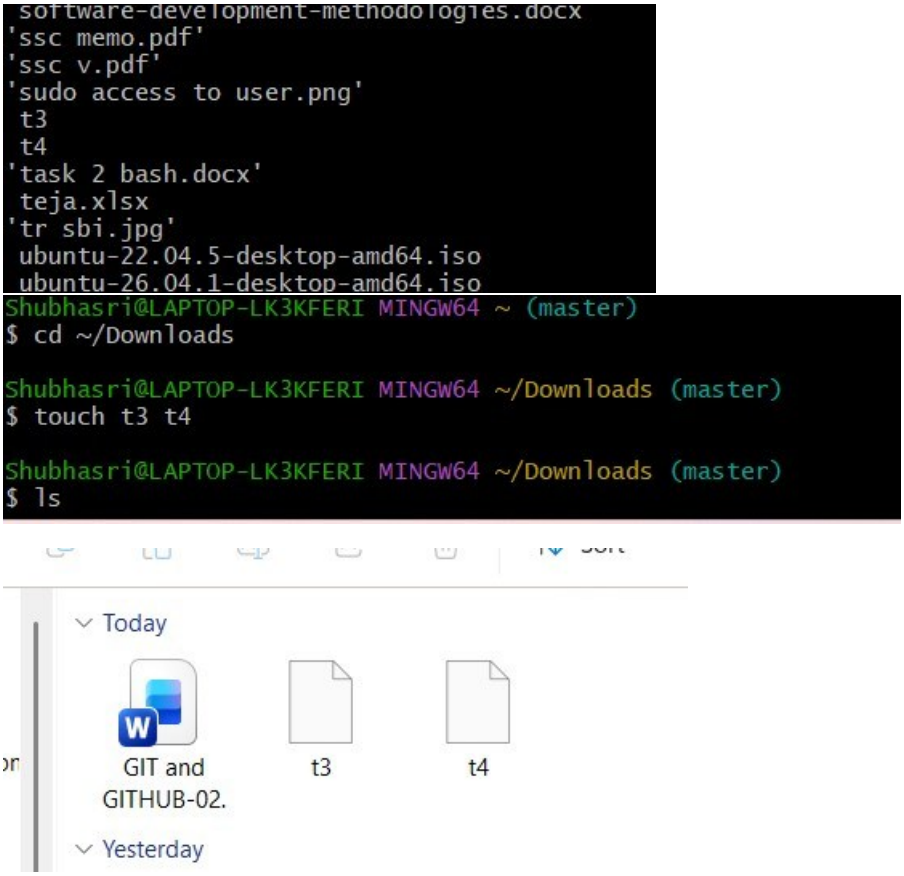
```
Shubhasri@LAPTOP-LK3KFERI MINGW64 /
$ cd ~

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~ (master)
$ cd ~/Desktop
bash: cd: /c/Users/Shubhasri/Desktop: No such file or directory

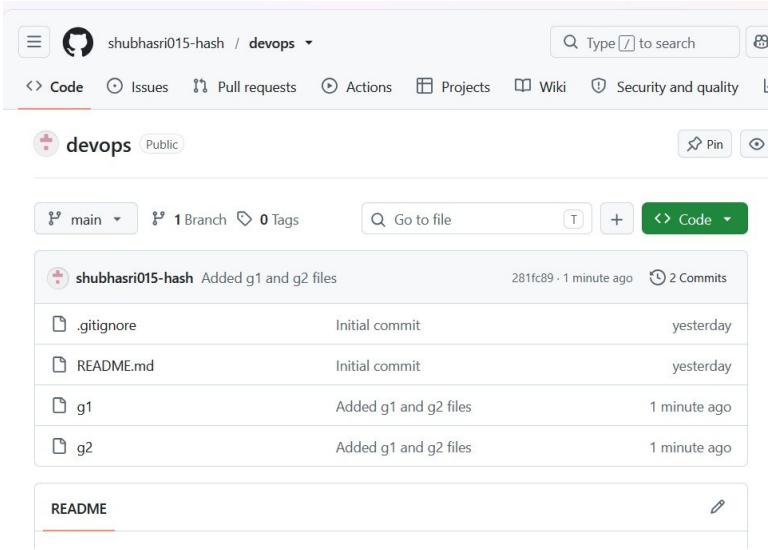
Shubhasri@LAPTOP-LK3KFERI MINGW64 ~ (master)
$ git clone git@github.com:shubhasri015-hash/task1.git
Cloning into 'task1'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 3 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
```

4.Create two files in local repo

To create files in local repo we need to use the command touch



5. Commit two files and push to central Repository



Create two files in local repo and add a message in that file by using echo command

And initialize git in that git config, Use git add . and git commit -m "Initial commit

Use use remote add orgin ssh – git branch -M main

Need to pull origin and the git push to origin main

```
$ git config --global user.name "shubhasri015-hash"

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads (master)
$ git config --global user.email "shubhasri015@gmail.com"

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads (master)
$ git add g1 g2

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads (master)
$ git commit -m "Added g1 and g2 files"
[master (root-commit) cd65db0] Added g1 and g2 files
 2 files changed, 2 insertions(+)
 create mode 100644 g1
 create mode 100644 g2

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads (master)
$ git remote add origin git@github.com:shubhasri015-hash/devops.git

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads (master)
$ git branch -M main

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads (main)
$ git push -u origin main
To github.com:shubhasri015-hash/devops.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:shubhasri015-hash/devops.git'
hint: Updates were rejected because the remote contains work that you do
hint: not have locally. This is usually caused by another repository pushing
hint: to the same ref. You may want to first integrate the remote changes
hint: (e.g., 'git pull ...') before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

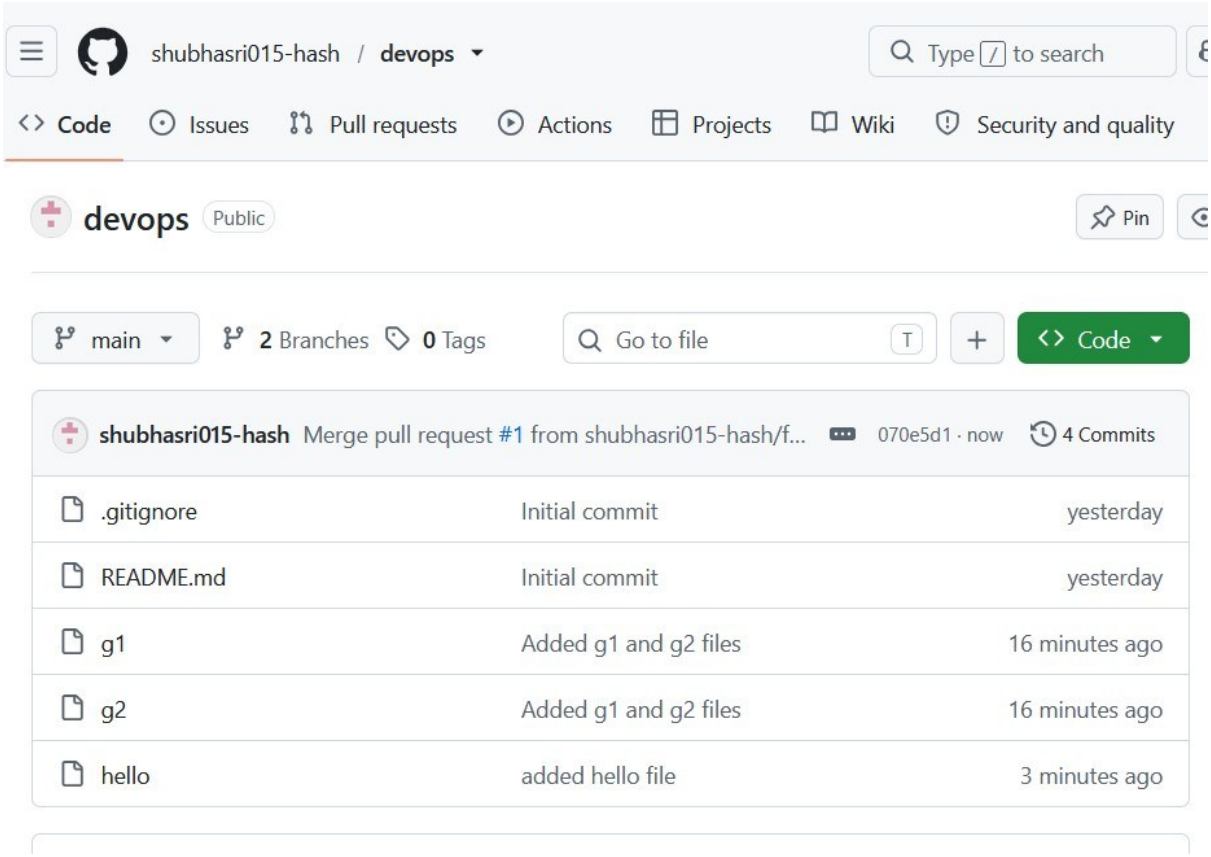
Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads (main)
$ git pull origin main --rebase
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 1.08 KiB | 11.00 KiB/s, done.
From github.com:shubhasri015-hash/devops
 * branch            main       -> FETCH_HEAD
 * [new branch]      main       -> origin/main
Successfully rebased and updated refs/heads/main.

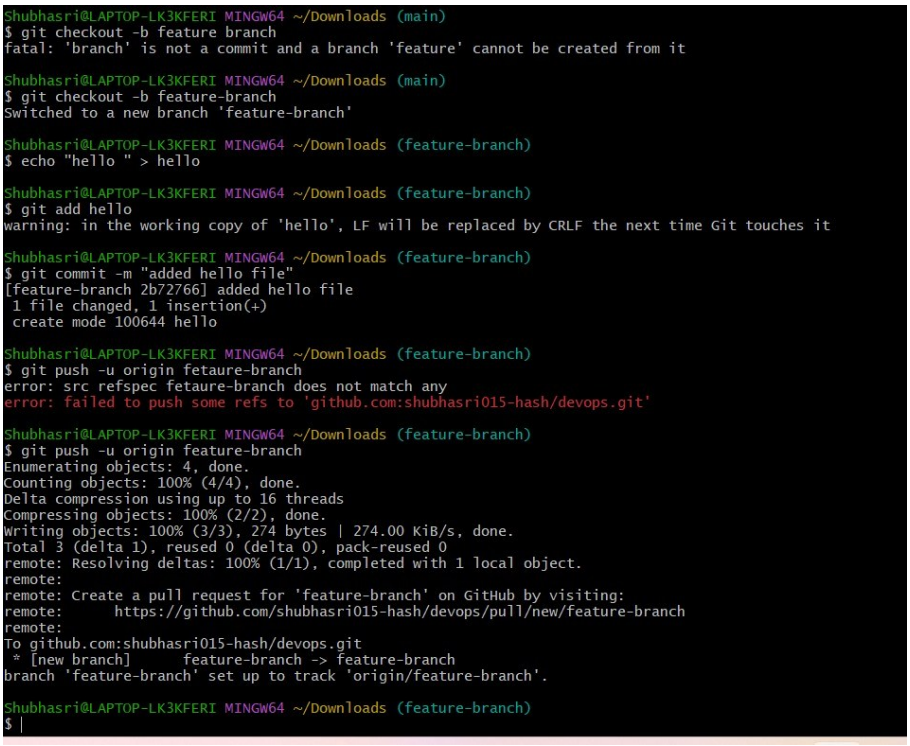
Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads (main)
$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 374 bytes | 374.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:shubhasri015-hash/devops.git
   4ce15cc..281fc89  main -> main
branch 'main' set up to track 'origin/main'.

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads (main)
```

6. Create a branch in local and create a sample file and push to central

Create a new branch. -Create a sample file. -Add content to the file.- Add and commit the file.- Push the branch to GitHub





7 . Create a branch in github and clone that to local

Create a branch in GitHub - Open Git Bash - Clone the repository- Go into the repository - Get the latest branches- Switch to the GitHub branch git clone — downloads the repository to your computer. git fetch origin — downloads information about remote branches. git checkout feature — switches to the feature branch locally

```
Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads
$ cd devops

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (main)
$ git clone git@github.com:shubhasri015-hash/devops.git
Cloning into 'devops'...
remote: Enumerating objects: 12, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (7/7), done.
Receiving objects: 100% (12/12), done.
remote: Total 12 (delta 3), reused 6 (delta 1), pack-reused 0 (from 0)
Resolving deltas: 100% (3/3), done.

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (main)
$ git fetch origin
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 8 (delta 2), reused 6 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (8/8), 1.34 KiB | 17.00 KiB/s, done.
From github.com:shubhasri015-hash/devops
   4ce15cc..070e5d1  main           -> origin/main
 * [new branch]      feature-branch -> origin/feature-branch

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (main)
$ git checkout release
error: pathspec 'release' did not match any file(s) known to git

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (main)
$ git checkout feature-branch
Switched to a new branch 'feature-branch'
branch 'feature-branch' set up to track 'origin/feature-branch'.

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$ git checkout -b release
Switched to a new branch 'release'

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (release)
$ git branch -a
  feature-branch
  main
* release
  remotes/origin/HEAD -> origin/main
  remotes/origin/feature-branch
  remotes/origin/main

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (release)
$ cd devops

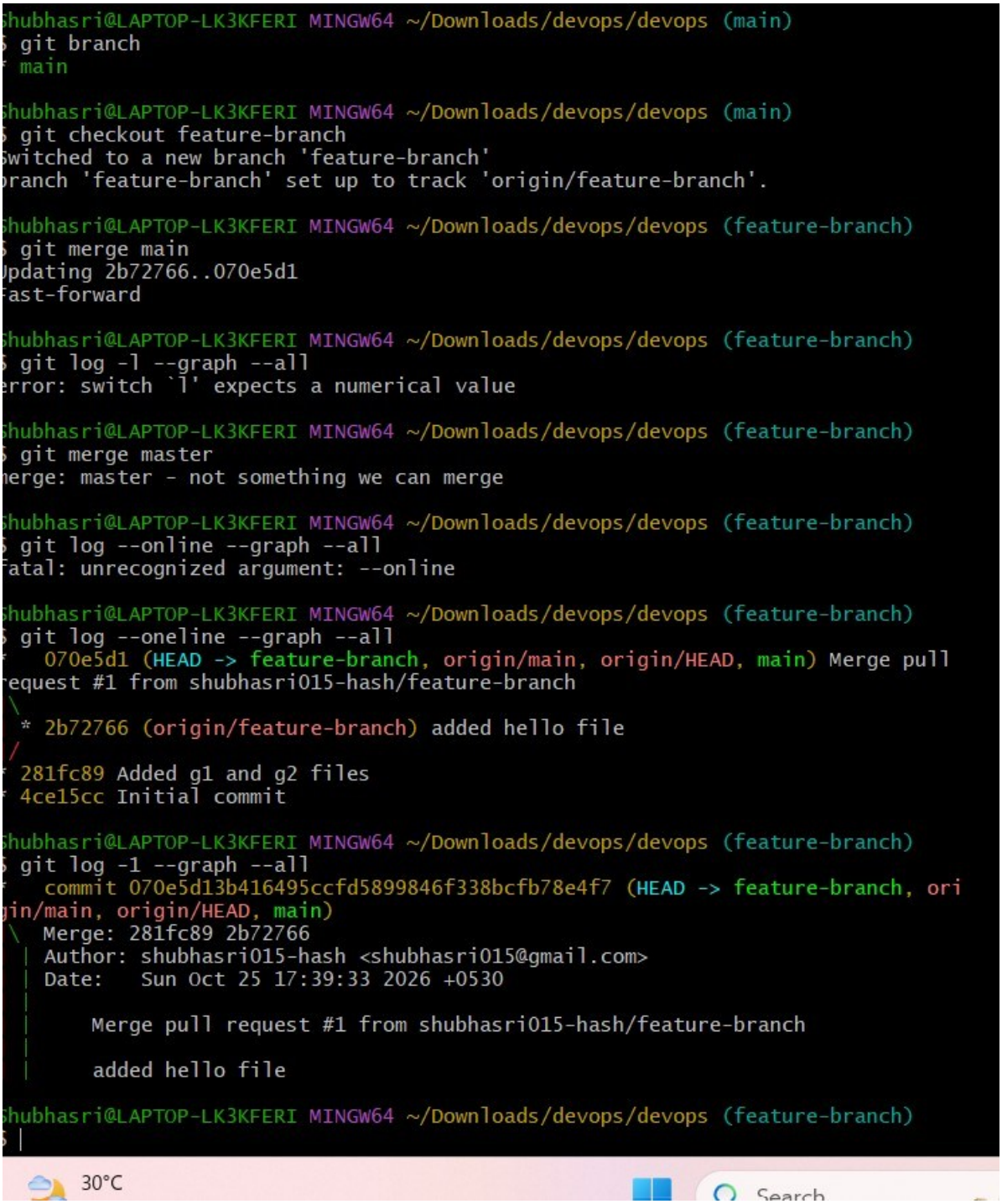
Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops/devops (main)
$ ls -a
./  ../  .git/  .gitignore  README.md  g1  g2  hello

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops/devops (main)
$
```

8. Merge the created branch with master in git local.

- git branch Lists local branches
- git switch master Switches to master

- git merge release Merges the created branch into master
- git log -1 --graph --all Displays branch and commit history
- git push origin master Uploads the merged changes to GitHub



9. Merge the created branch with master in github by sending a pull request.

Create a branch in your GitHub repository, feature -branch .Make changes to a file in the feature branch.Commit and push the changes to GitHub.Open your GitHub repository in the browser.Click the Pull requests tab.

Click New pull request.Select the branches:Base: main — the branch that receives the changes.Compare: release — the branch containing your changes.Click Create pull request.Enter a title and description, then click Create pull request.Review the changes and click Merge pull request.-Click Confirm merge to complete the process.

```
Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops/devops (feature-branch)
$ echo "This is my feature-branch" > hello

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops/devops (feature-branch)
$ git status
On branch feature-branch
Your branch is ahead of 'origin/feature-branch' by 1 commit.
  (use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   hello

no changes added to commit (use "git add" and/or "git commit -a")

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops/devops (feature-branch)
$ git add .
warning: in the working copy of 'hello', LF will be replaced by CRLF the next ti
me Git touches it

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops/devops (feature-branch)
$ git add hello

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops/devops (feature-branch)
$ git commit -m "added hello to feature-branch"
[feature-branch 201a8a3] added hello to feature-branch
 1 file changed, 1 insertion(+), 1 deletion(-)

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops/devops (feature-branch)
$ git push -u origin feature-branch
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 289 bytes | 289.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:shubhasri015-hash/devops.git
   2b72766..201a8a3  feature-branch -> feature-branch
branch 'feature-branch' set up to track 'origin/feature-branch'.

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops/devops (feature-branch)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops/devops (main)
$ git pull origin main
From github.com:shubhasri015-hash/devops
 * branch            main       -> FETCH_HEAD
Already up to date.

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops/devops (main)
$ ls
README.md  g1  g2  hello
```

10. Create a file in local and send that to branch in github.

- Switch to release branch
- Create sample file
- Read file cat sample.txt
- Git add

- Git commit –m
- Git push –u origin release

```
hello

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops/devops (main)
$ git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops/devops (main)
$ echo "this is github file" > hello

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops/devops (main)
$ ls
README.md  g1  g2  hello

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops/devops (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   hello

no changes added to commit (use "git add" and/or "git commit -a")

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops/devops (main)
$ git add hello
warning: in the working copy of 'hello', LF will be replaced by CRLF the next ti
me Git touches it

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops/devops (main)
$ git commit -m "added file"
[main ab39fe6] added file
 1 file changed, 1 insertion(+), 1 deletion(-)

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops/devops (main)
$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 270 bytes | 270.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:shubhasri015-hash/devops.git
   070e5d1..ab39fe6  main -> main
branch 'main' set up to track 'origin/main'.

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops/devops (main)
$
```

11. Clone only a branch from github to local

- Clone a specific branch REPOSITORY_URL — git clone --branch feature-branch --single-branch
- Enter repository — cd my-repository
- Check current branch — git branch
- Check remote branches — git branch -r

```
Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads (master)
$ cd devops
bash: cd: devops: No such file or directory

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads (master)
$ git clone --branch feature-branch --single-branch git@github.com:shubhasri015-
hash/devops.git
Cloning into 'devops'...
remote: Enumerating objects: 15, done.
remote: Counting objects: 100% (15/15), done.
remote: Compressing objects: 100% (9/9), done.
remote: Total 15 (delta 4), reused 8 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (15/15), done.
Resolving deltas: 100% (4/4), done.

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads (master)
$ cd devops

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$ git branch
* feature-branch

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$ git branch -r
  origin/feature-branch

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$
```

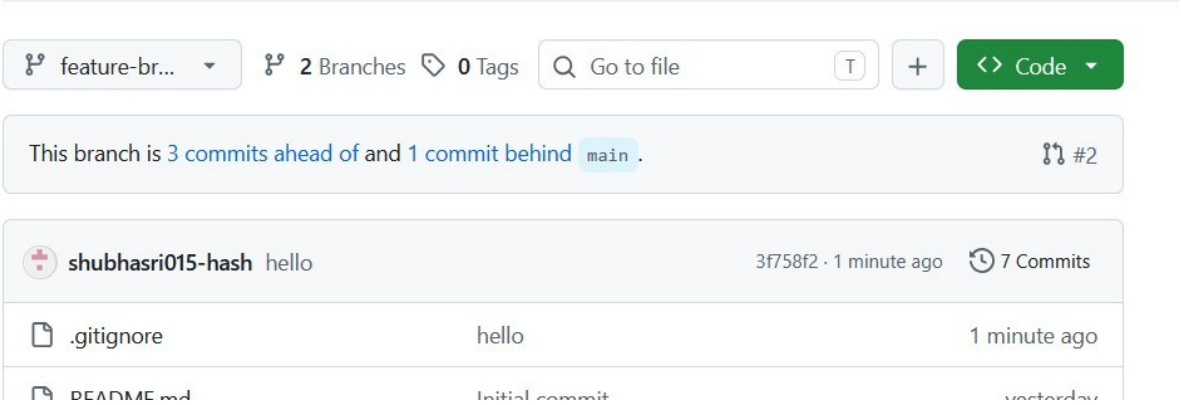
12. Create a file with all passwords and make that untrackable with git

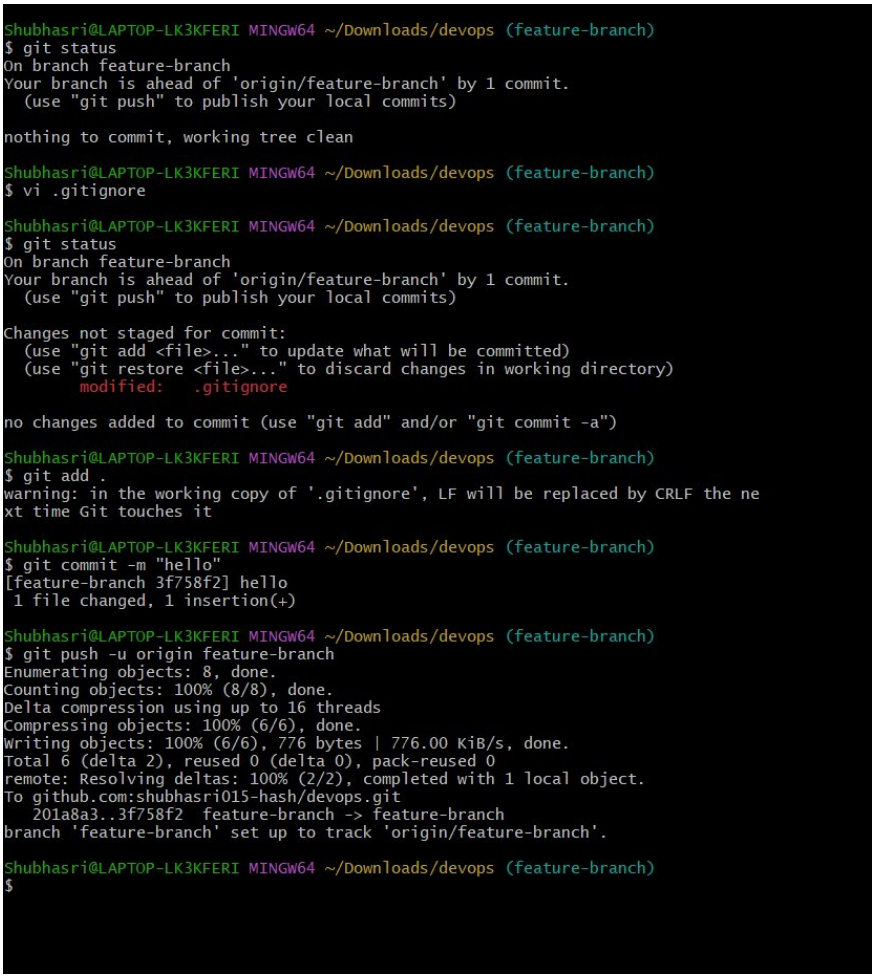
Vi to edit the file.

gitignore tells Git not to track matching files.

git status checks whether Git is tracking the file.

git commit saves the .gitignore change





13. Make a commit and make that commit reset without savings changes.

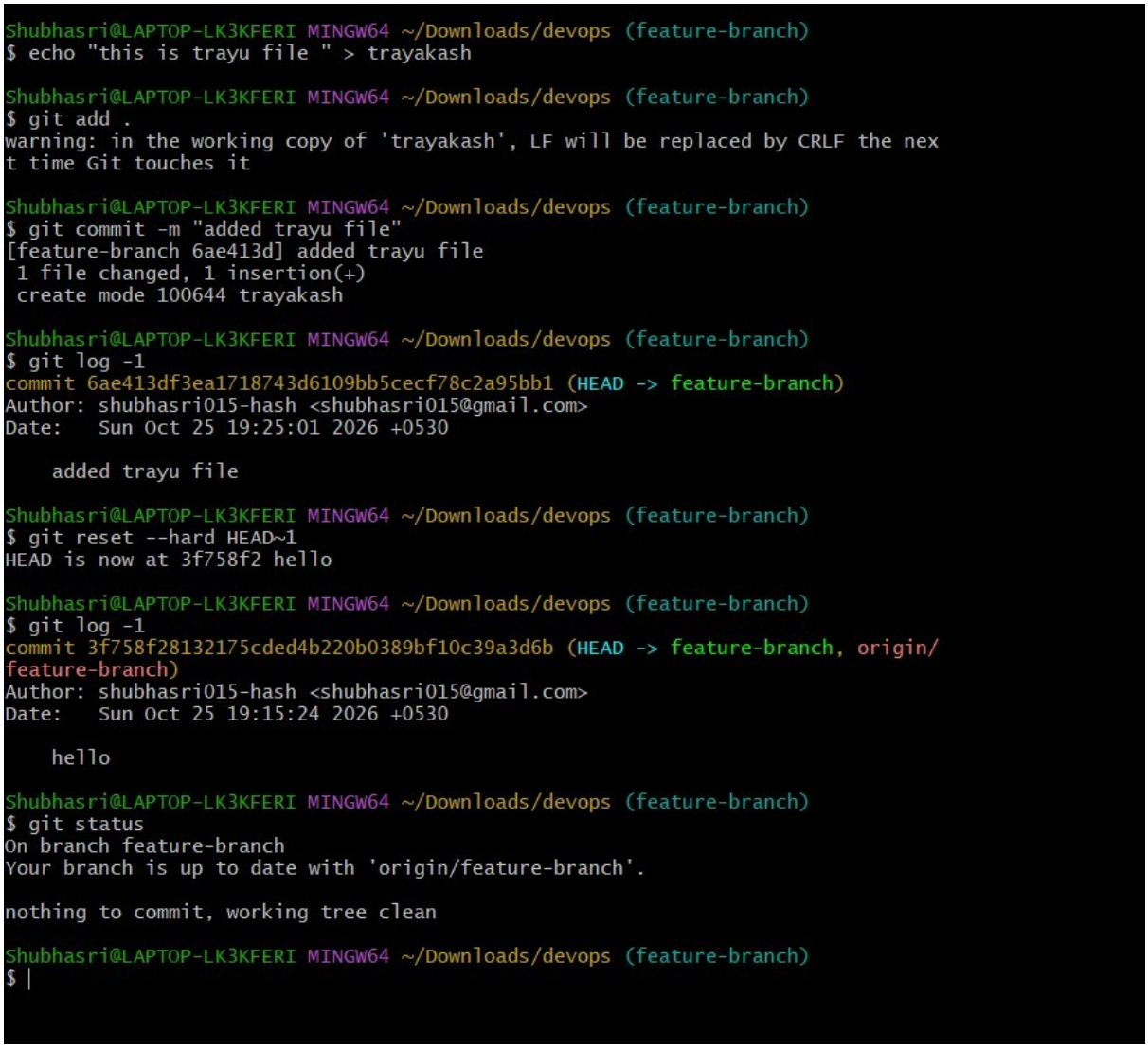
echo "this is trayu file" > trayaksh – Creates a file with text.

git add .– Stages the file.

git commit -m "Added trayu file " – Saves the changes in a commit.

git reset --hard HEAD~1 – Removes the latest local commit and discards its changes.

git status – Checks the current repository status.



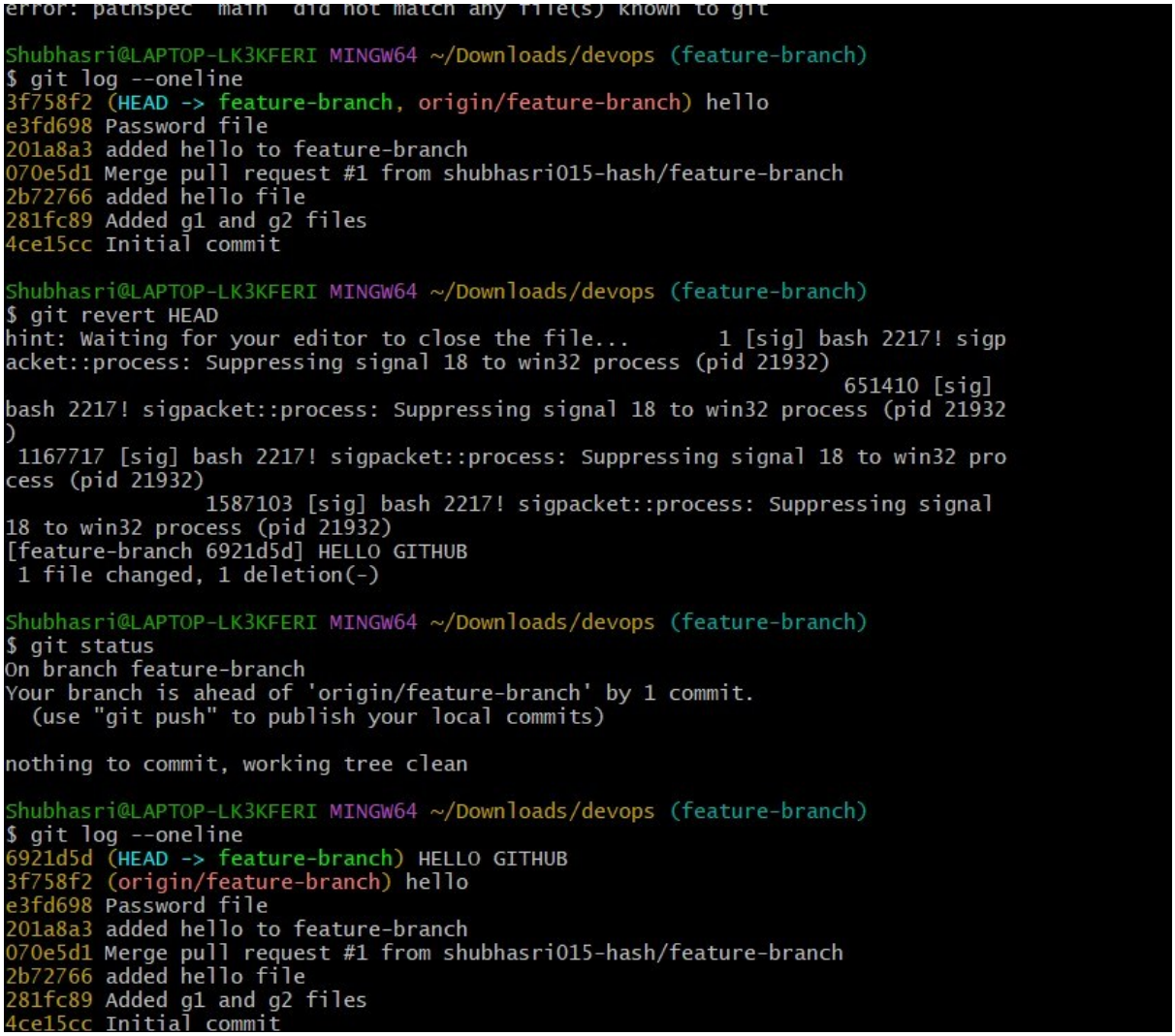
14. Revert a commited commit to the older version.

git log --oneline – Displays the commit history. git revert HEAD –

Undoes the latest commit by creating a new commit.

git status – Checks the repository status.

git log --oneline – Verifies the new revert commit.



15. Push a file to stash without savings the changes and work on another file.

git stash – Temporarily saves changes without committing them.

git status – Checks the repository status. git

stash pop – Restores the stashed changes.

```
Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$ ls
README.md  g1  g2  hello  kite  password.txt  tiger

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$ vi goat

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$ git staus
git: 'staus' is not a git command. See 'git --help'.

The most similar command is
        status

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$ git status
On branch feature-branch
Your branch is ahead of 'origin/feature-branch' by 1 commit.
  (use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   hello

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        goat
        kite
        tiger

no changes added to commit (use "git add" and/or "git commit -a")

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$ git add .
warning: in the working copy of 'goat', LF will be replaced by CRLF the next tim
e Git touches it
warning: in the working copy of 'kite', LF will be replaced by CRLF the next tim
e Git touches it
warning: in the working copy of 'tiger', LF will be replaced by CRLF the next ti
me Git touches it

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$ git status
On branch feature-branch
Your branch is ahead of 'origin/feature-branch' by 1 commit.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   goat
        modified:   hello
        new file:   kite
        new file:   tiger


Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
```

```
Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$ ls
README.md  g1  g2  goat  hello  kite  password.txt  tiger

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$ git stash
Saved working directory and index state WIP on feature-branch: 6921d5d HELLO GIT
HUB

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$ vi donkey

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$ git add .
warning: in the working copy of 'donkey', LF will be replaced by CRLF the next t
ime Git touches it

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$ git commit -m "donkey"
[feature-branch f110d6a] donkey
 1 file changed, 1 insertion(+)
 create mode 100644 donkey

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$ ls
README.md  donkey  g1  g2  hello  password.txt

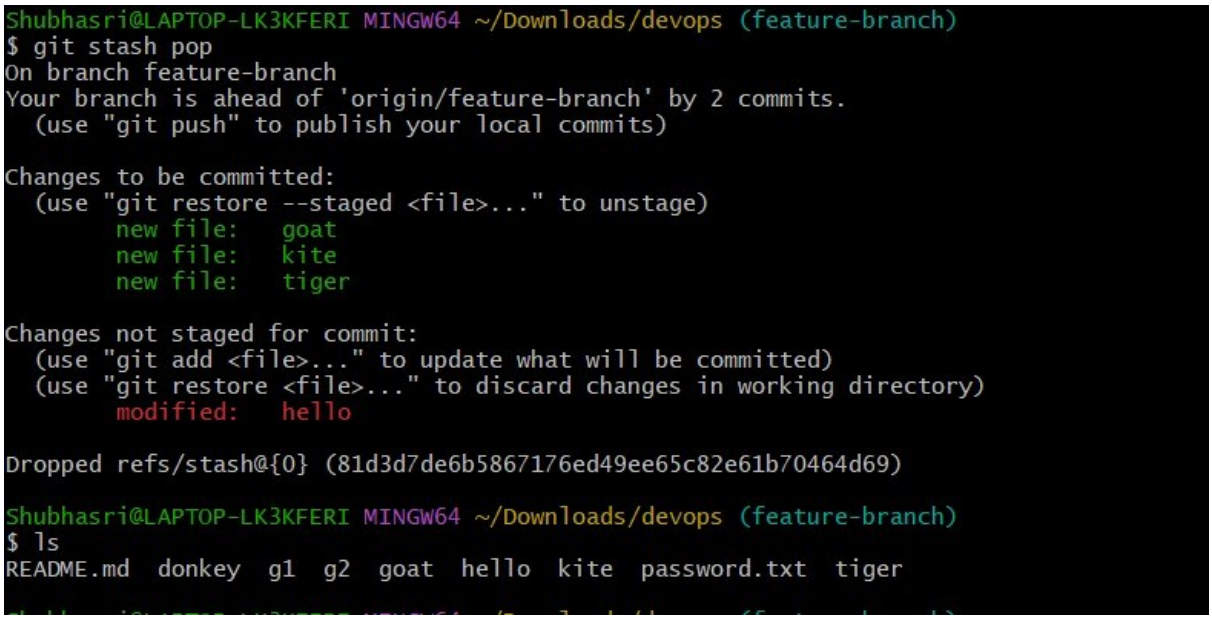
Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$ git stash list
stash@{0}: WIP on feature-branch: 6921d5d HELLO GITHUB

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$
```

16.

Undo the stash file and start working on that again

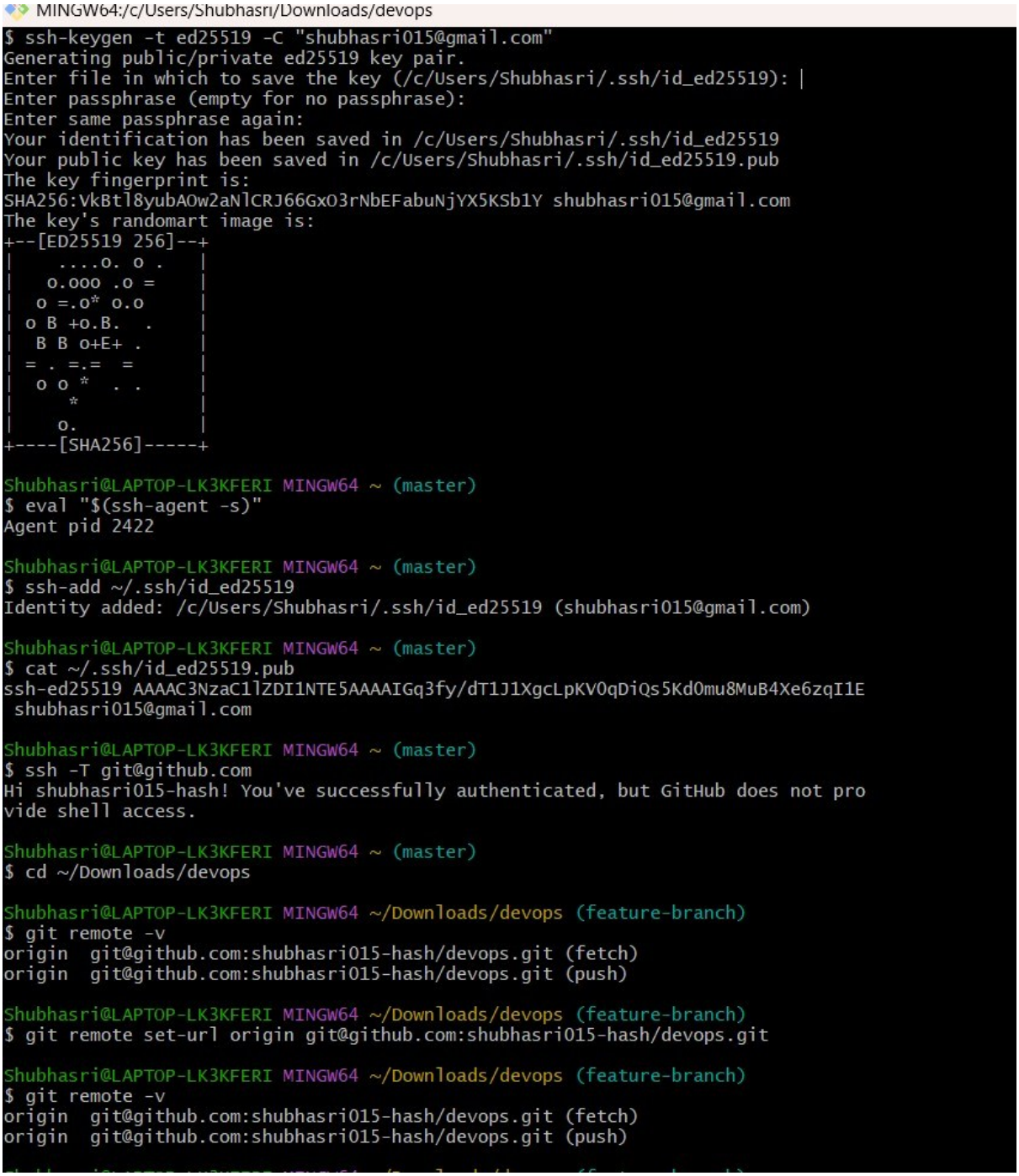
git stash pop is used to restore the changes saved in the stash and remove them from the stash list.



17.

Generate a ssh-keygen and configure into github.

- **Generate key: Create your SSH key pair.**
- **Start agent: Start the SSH agent.**
- **Add key: Load your private key into the agent.**
- **Display public key: Copy the public key and add it under GitHub → Settings → SSH and GPG keys.**
- **Test connection: Run ssh -T git@github.com.**
- **Check remote: Run git remote -v to check your repository URL. • Fetch changes: Run git fetch origin to retrieve remote updates.**



18.

Configure webhooks to github.

Open your GitHub repository.

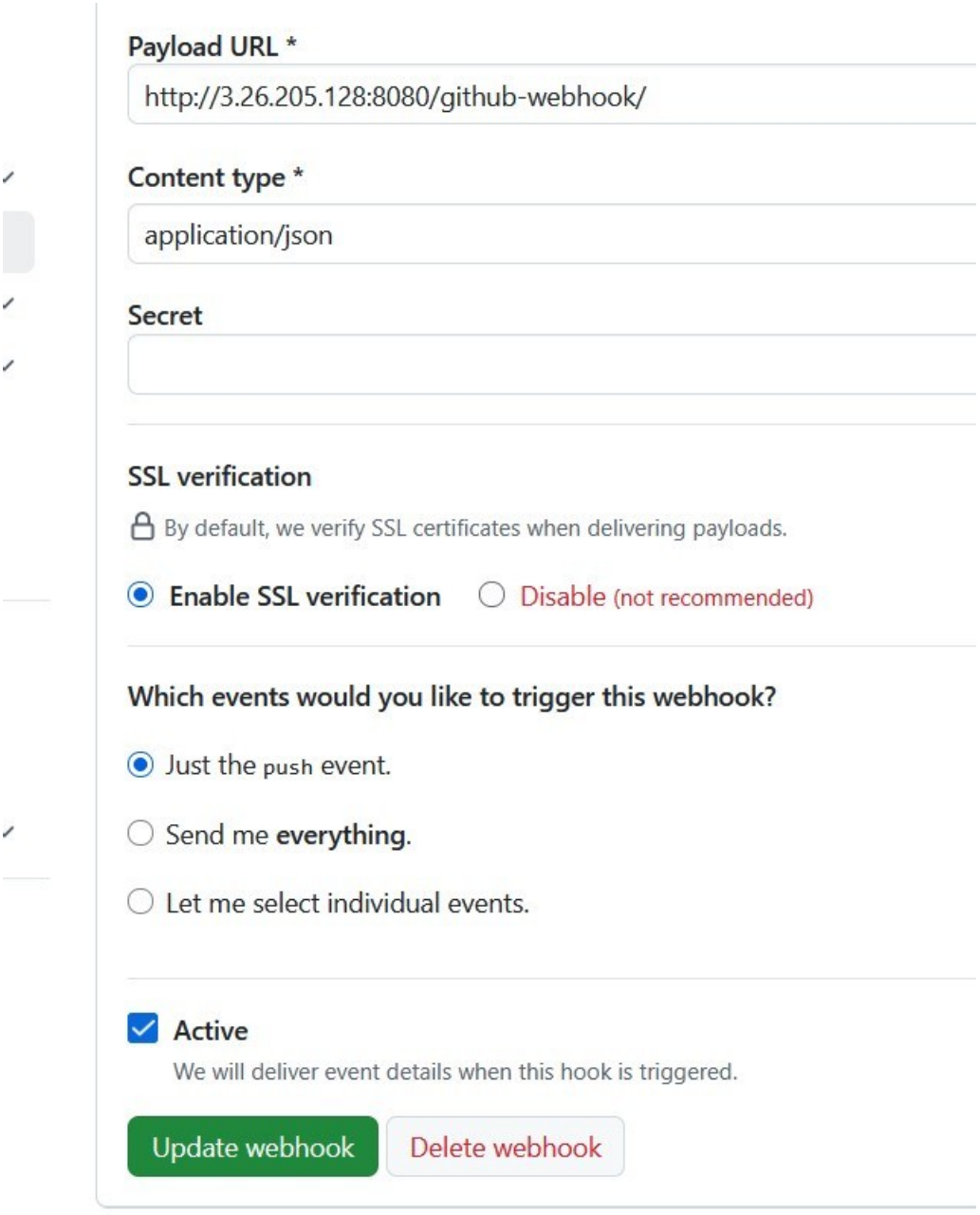
Go to Settings → Webhooks → Add webhook.

Enter the Jenkins webhook URL in the Payload URL field.

Select application/json.

Choose Just the push event.

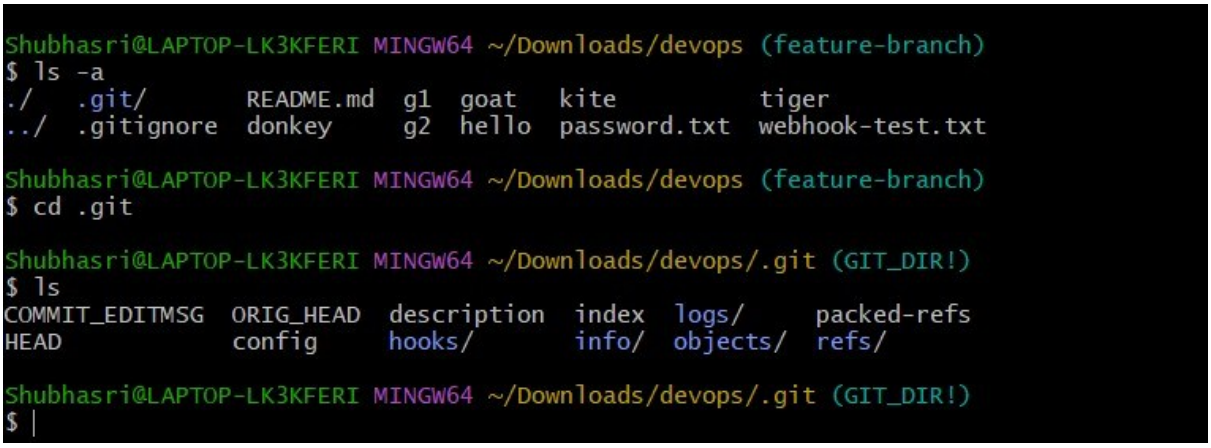
Click Add webhook.



19.

Basic understanding of .git directory

The .git directory is the core of a local Git repository. If you delete it, your project files remain, but the repository's Git history and metadata are lost.



Displays Git's internal files and folders.

Important contents

File/Directory	Purpose
HEAD	**Identifies the current branch or commit reference.**
config	**Stores repository-specific Git settings.**
objects/	**Stores Git objects, including file contents and commits.**
refs/	**Stores references to branches and tags.**
hooks/	**Contains sample or configured scripts that run during Git events.**
index	**Stores the staging area's information.**

20. Check all the logs of git.

- **Use git log to check the commit history of your local repository.**
  Git log Commit ID
  Author name
  Date and time
  Commit message

```
Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$ git log
commit 74cbbb215eefa270ab2762d4cb48b56c41d8ce55 (HEAD -> feature-branch, origin/
feature-branch)
Author: shubhasri015-hash <shubhasri015@gmail.com>
Date:   Sun Oct 25 20:18:30 2026 +0530

    test github webhook

commit f110d6a9b659f741fc0ce0ced4403db69173b21c
Author: shubhasri015-hash <shubhasri015@gmail.com>
Date:   Sun Oct 25 19:53:33 2026 +0530

    donkey

commit 6921d5d57c18df3a18e7a56557cfdfd0b59ad7d8
Author: shubhasri015-hash <shubhasri015@gmail.com>
```

```
Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$ git log --oneline
74cbbb2 (HEAD -> feature-branch, origin/feature-branch) test github webhook
f110d6a donkey
6921d5d HELLO GITHUB
3f758f2 hello
e3fd698 Password file
201a8a3 added hello to feature-branch
070e5d1 Merge pull request #1 from shubhasri015-hash/feature-branch
2b72766 added hello file
281fc89 Added g1 and g2 files
4ce15cc Initial commit

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$ git log --all
commit 74cbbb215eefa270ab2762d4cb48b56c41d8ce55 (HEAD -> feature-branch, origin/
feature-branch)
Author: shubhasri015-hash <shubhasri015@gmail.com>
Date:   Sun Oct 25 20:18:30 2026 +0530

    test github webhook
```

```
    Initial commit

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Downloads/devops (feature-branch)
$ git log --oneline --all --graph
* 74cbbb2 (HEAD -> feature-branch, origin/feature-branch) test github webhook
* f110d6a donkey
* 6921d5d HELLO GITHUB
* 3f758f2 hello
* e3fd698 Password file
* 201a8a3 added hello to feature-branch
*   070e5d1 Merge pull request #1 from shubhasri015-hash/feature-branch
|\
| * 2b72766 added hello file
|/
* 281fc89 Added g1 and g2 files
* 4ce15cc Initial commit
```

git log --oneline --all –graph

Shows commit history and branch merges in a graphical format.

git log –all

Displays commit history from all locally known branches.

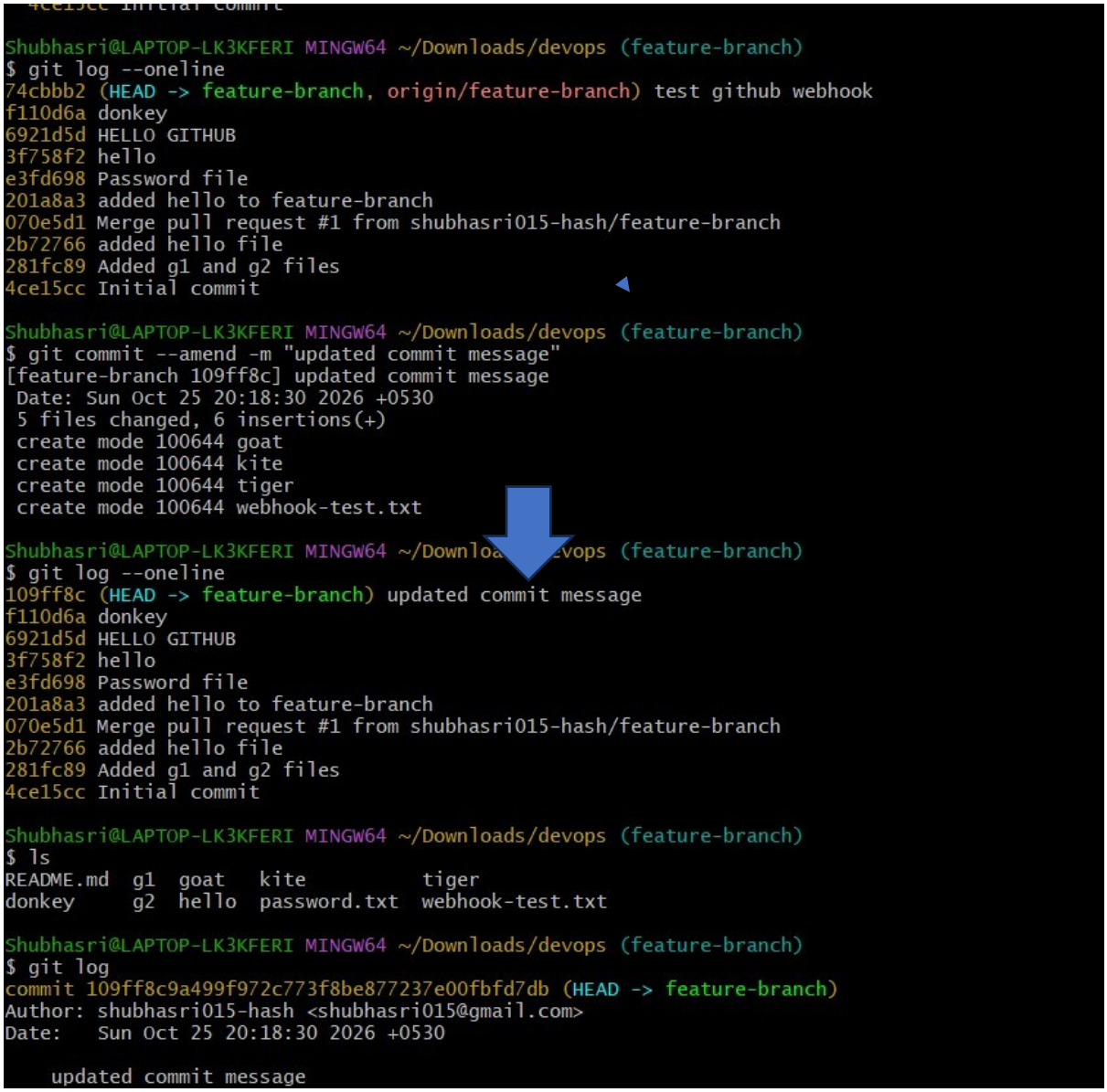
git log –oneline

Shows a short summary of each commit.

21. Rename the commit message.

git commit --amend -m is used to modify the latest commit message.

Note: If the commit was already pushed to GitHub, updating it changes its commit ID. You may need to push with git push --force-with-lease; use caution if others share the branch.



22. Merge multiple commits into single commit

Check commit history - Start interactive rebase- Enter one message for the combined commit, then save and close the editor git rebase -i lets you combine multiple commits into a single commit using squash.

Note: This rewrites commit history. If those commits were already pushed to GitHub, coordinate with collaborators before pushing the rewritten history.

